THE KARNATAKA LAW SOCIETY'S
GOGTE INSTITUTE OF TECHNOLOGY, BELAGAVI
(An Autonomous Institution under VTU, Accredited by NAAC A+)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Open Book Assignment I

Course Name	Digital Design and Computer Organization	**Course Code**	1BCS305
Semester	III	**Division**	A, B, C, D, E
Date	13/10/2026	**Maximum Marks**	10

Instructions:

1. ***All questions are compulsory.***
2. ***Show the necessary steps used in solving the problems, such as truth tables, simplification etc…***
3. ***Take screenshots of the circuit and simulation in LogiSim and attach in the pink book.***

1. Consider the following Boolean expression:

$$Y = \bar{A} \cdot \bar{B} + A \cdot B$$

a) Draw the truth table for this expression.

b) Determine the operation being performed.

c) Realize the expression using Basic Gates.

d) Realize the expression using only NOR Gates. (Hint: Optimize to use only 4 NOR Gates)

e) Implement the circuits in (c) and (d) using LogiSim.

3	1,2,3	1,3,5	5
[L]	**[CO]**	**[PO]**	**[M]**

2. Perform the following number conversions:

a) 2025.9375 from decimal to binary and hexadecimal

b) 101111011.101011 from binary to decimal and hexadecimal

c) FCDB.EA19 from hexadecimal to decimal and binary

3	1,2	1	3
[L]	**[CO]**	**[PO]**	**[M]**

3. Perform the following arithmetic operations using 2's Complement arithmetic:

a) 3AFCD – 2BDE9

b) $(-32768)_{10} + (19865)_{10}$

3	1,2	1	2
[L]	**[CO]**	**[PO]**	**[M]**

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Signature of the Faculty Members	**Signature of the Module Coordinator**	**HOD**